

EXP 4

Code:

```
transmit_power=30;

frequency=900e6;

antenna_height_t=40;

antenna_height_r=10;

d=1:1:1000;

free_space_path_loss=(20*log10(d))+20*log10(frequency)-147.55;

two_ray_ground_path_loss=(40*log10(d))+20*log10(antenna_height_t)+log10(antenna_height_r)+20*log10(frequency)-147.55;

figure;

plot(d,free_space_path_loss,'b','DisplayName','Free space path loss');

hold on;

plot(d,two_ray_ground_path_loss,'r','DisplayName','Two ray ground path loss');

xlabel('Distance(m)');

ylabel('Path loss(dB)');

title('Effect of propagation model on coverage distance');

legend;

grid on;

hold off;
```

Output :

